Revision 2021 • Semester 2 • Engineering Science Lab

Engineering Mechanics Lab

A professional hands-on handbook for Diploma course 2029. It converts the official lab syllabus into experiment chapters, apparatus guides, vector diagrams, observation formats, formula banks, solved calculations, viva preparation, troubleshooting notes and a fresh practical model paper.

2029	**45**	**No Credit**

Force system to field judgement

Draw
force polygon, beam diagram, centroid lines, friction plane.

Calculate
resultant, reactions, centroid, moment of inertia, strain, coefficient of friction.

Test
apparatus alignment, readings, repeatability, percentage error and conclusion.

Course Overview

Engineering Mechanics Lab is the practical bridge between theory and real structures or machines. Students do not just memorize formulas; they set up apparatus, observe equilibrium, measure geometry, calculate unknowns, compare error and write a defensible lab conclusion.

What the subject is

A laboratory course covering force systems, support reactions, centroid, mass moment of inertia, strain under tensile load and coefficient of friction. The official syllabus lists it as a second-semester Engineering Science laboratory.

Industrial relevance

These experiments are used in beam support design, lifting frames, jigs, cranes, machine shafts, braking, guide rail contact, escalator/elevator structures and general maintenance troubleshooting.

How to study

For each experiment learn five items in order: aim, apparatus, diagram, observation table, calculation and conclusion. Weak students usually fail because they skip units, scale and error analysis.

Official objectives

- Provide hands-on experience to outline basic concepts of engineering mechanics.
- Experiment with force systems and use them to solve engineering problems.

□ lab-□ theory □□□□□□ □□□□. Apparatus □□□□□□□ set □□□□□□□, reading □□□□□□□□, formula substitute □□□□□□□, result unit □□□□□ □□□□□□.

Prerequisites

- Knowledge of basic Mathematics from Mathematics I, Semester 1.
- Knowledge of basic Physics from Applied Physics I, Semester 1.

Meaning: trigonometry, scale drawing, moment equation, unit conversion and basic force concepts must be clear before entering the lab.

Source / syllabus information

Item	Confirmed official information	Missing / limitation
Course code and title	2029 - Engineering Mechanics Lab	None
Programme	Architecture, Automobile, Civil and allied, Mechanical and allied, Wood & Paper Technology, Environmental Engineering	Exact department-wise exam paper pattern not supplied.
Semester / category	Semester 2 / Engineering Science	None
Credits and periods	No Credit; 3 periods/week; 45 periods/semester	Detailed mark distribution not specified in supplied syllabus/source.
CO-PO mapping	CO1 to CO4 are strongly mapped to PO4 with value 3.	PO descriptions are not included in the supplied source.
Model QP pattern	Lab Exam I and Lab Exam II are mentioned.	Official model practical question paper pattern not specified in supplied syllabus/source. The model paper here is freshly created for preparation.

Course outcomes and engineering meaning

CO	Official outcome	Hours	Engineering meaning
CO1	Identify force systems for given conditions by applying the basics of mechanics.	15	Recognise concurrent, parallel and crane-like force systems and find/resultant or equilibrant using apparatus and drawing.
CO2	Determine unknown forces of different engineering systems.	6	Find beam support reactions analytically and graphically.
CO3	Infer centre of gravity and mass moment of inertia.	9	Locate centroid of laminae and determine flywheel inertia for rotating systems.
CO4	Determine strains in mutually perpendicular directions under axial tension. Determine coefficient of friction on a plane through experimentation.	11	Measure deformation and friction behaviour using laboratory equipment and interpret engineering safety/application.

Table of Contents

Every button below opens a real section without creating browser-back history stacking.

Redesigned Syllabus Roadmap

This roadmap converts the official lab list into what to set, draw, calculate, explain and avoid in examination.

CO1 • 15h

Force systems and Jib crane

Core topics: lab apparatus, concurrent resultant by force table, graphical polygon, parallel forces, Lami's theorem.

Draw: force table layout, vector polygon, space diagram, Jib crane member force triangle.

Calculate: resultant/equilibrant magnitude and direction. **Common mistake:** treating equilibrant as resultant without reversing direction.

CO2 • 6h + Lab I

Support reactions

Core topics: simply supported beam reactions and link polygon method.

Draw: beam diagram, load positions, reaction arrows, funicular/link polygon.

Calculate: R_A, R_B from $\Sigma V=0$ and $\Sigma M=0$. **Common mistake:** taking wrong distance from support.

CO3 • 9h

Centroid and mass moment of inertia

Core topics: centroid of laminae, flywheel and shaft inertia, bell crank lever.

Draw: lamina suspension lines, flywheel string arrangement, lever arms.

Calculate: centroid location, moment balance, inertia terms. **Common mistake:** confusing centroid with centre of gravity for non-uniform bodies.

CO4 • 11h + Lab II

Strain and friction

Core topics: strain under tensile loading, friction on horizontal and inclined planes, open-ended experiment.

Draw: tensile specimen, load-extension graph, friction block, inclined plane.

Calculate: strain, $\mu=F/N$, $\mu=\tan\theta$. **Common mistake:** mixing gram force, newton and kilogram mass.

Practical/field applications

- Beam support reactions: machine base frames, staircase beams, scaffold planks and escalator truss support checks.
- Friction: brake pads, belt drives, guide shoes, material handling chutes and sliding machine parts.
- Moment of inertia: flywheels, shafts, rotating machine components and balancing.
- Strain: quality inspection of material under load and structural safety judgement.

Lab safety and discipline

- Never place fingers under suspended weights or moving lever arms.

- Check zero error before reading spring balance or gauge.
- Keep pulleys clean and aligned; pulley friction creates major error.
- Record unit with every reading. Unitless lab records are not engineering records.

Weight □□□□□□□□ jerk □□□□□□□□□□□. Reading stable □□□□□□ □□□□ □□□□□□ note □□□□□□□.

Module 1 • CO1 • 15 Hours

Force Systems, Resultant and Lami’s Theorem

Outcome: identify force systems for given conditions by applying mechanics basics. This module is where students learn how a group of forces becomes one equivalent resultant and how equilibrium is verified in a real apparatus.

M1.01 Identify apparatus related to Engineering Mechanics

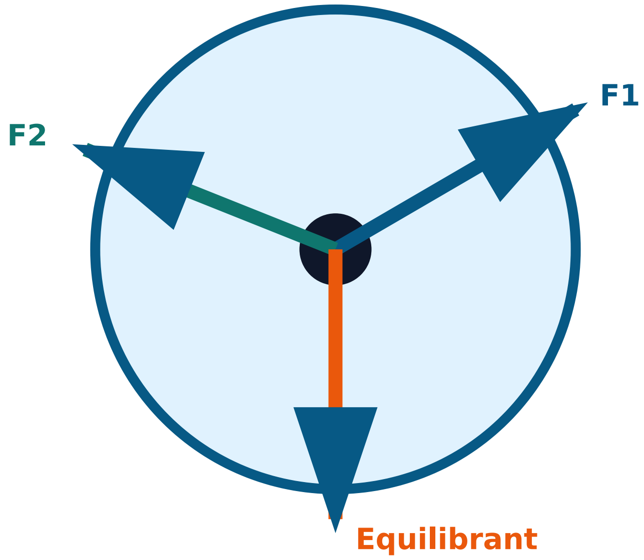
Common apparatus include force table, pulleys, slotted weights, hangers, spring balance, drawing board, set square, protractor, beam apparatus, plumb bob, lamina, flywheel apparatus, bell crank lever, tensile testing arrangement and friction plane. Identification is not just naming; a student must explain purpose, reading method, possible zero error and safety issue.

Apparatus name □□□□□□ □□□□. □□□ □□□□□□□ □□□□□□□□□□□□□□, reading □□□□□□ □□□□□□□□□□□, error □□□□□ □□□□ □□□□□□ exam-□ □□□□□.

Exam answer format

1. Name the apparatus.
2. Write one use.
3. State one precaution.
4. Connect it to the experiment aim.

Force table concept



M1.02 Resultant of concurrent force system using force table

Concurrent forces pass through one common point. In force-table experiment, known weights act through strings over pulleys at selected angles. The centre ring must remain exactly at the table centre. The balancing force is the equilibrant; the resultant is equal in magnitude and opposite in direction to the equilibrant.

Procedure

1. Level the force table and check pulley smoothness.
2. Place pulleys at required angles and hang known weights.
3. Add a third hanger and adjust angle/weight until the ring is centered.
4. Record force magnitudes and angles.
5. Draw vector polygon and compare calculated resultant with experimental equilibrant.

M1.03 & M1.04 Graphical resultant of concurrent and parallel forces

For concurrent forces, draw force vectors head-to-tail using a selected scale. The closing side gives resultant. For parallel forces, draw a space diagram and use a force polygon/link polygon to find resultant location. The quality of result depends on accurate scale, arrow direction and angle measurement.

Scale rule: if 1 cm = 10 N and measured closing side is 5.2 cm, resultant = 52 N. Direction must be read from the drawing, not guessed.

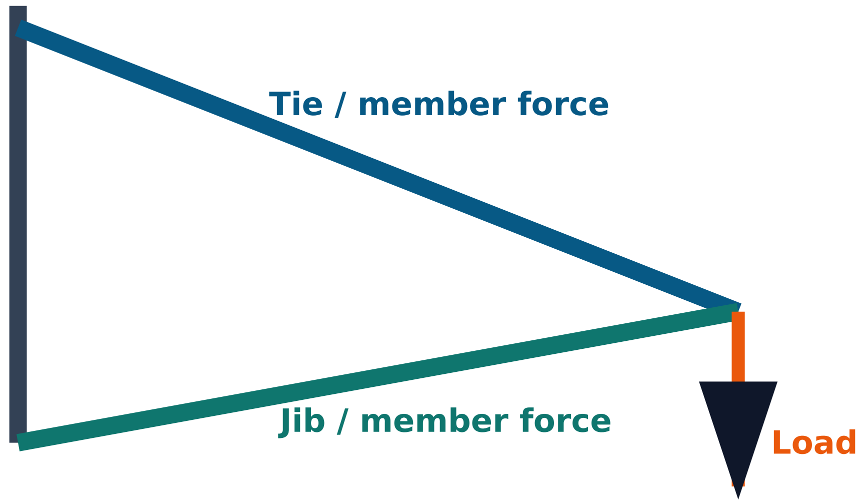
M1.05 Verify Lami’s theorem using Jib crane

Lami's theorem applies when three coplanar concurrent forces keep a body in equilibrium. For the Jib crane, the three forces are load, force in tie and force in jib. If the angles opposite to the forces are α, β and γ:

$$F1 / \sin \alpha = F2 / \sin \beta = F3 / \sin \gamma$$

The experiment confirms that member forces calculated from the force triangle match observed forces within acceptable error.

Jib crane force triangle



Common mistakes

- Not keeping ring exactly at centre.
- Ignoring hanger weight when it is significant.
- Using wrong protractor direction.
- Writing equilibrant as resultant without reversing 180°.

Module 1 expected questions

- List apparatus used in force table experiment.
- Explain resultant and equilibrant.
- State and verify Lami's theorem.
- Draw vector polygon for three forces.
- Explain errors in graphical method.

Module 2 • CO2 • 6 Hours + Lab Exam I

Support Reactions of Beams

Outcome: determine unknown forces of engineering systems. Beam reactions are basic for structures, machines, stairs, platforms and escalator support framing.

M2.01 Determine support reactions for simply supported beam

A simply supported beam usually has a pin support and a roller support. Downward loads are balanced by upward reactions at supports. For vertical loading, use two equilibrium equations:

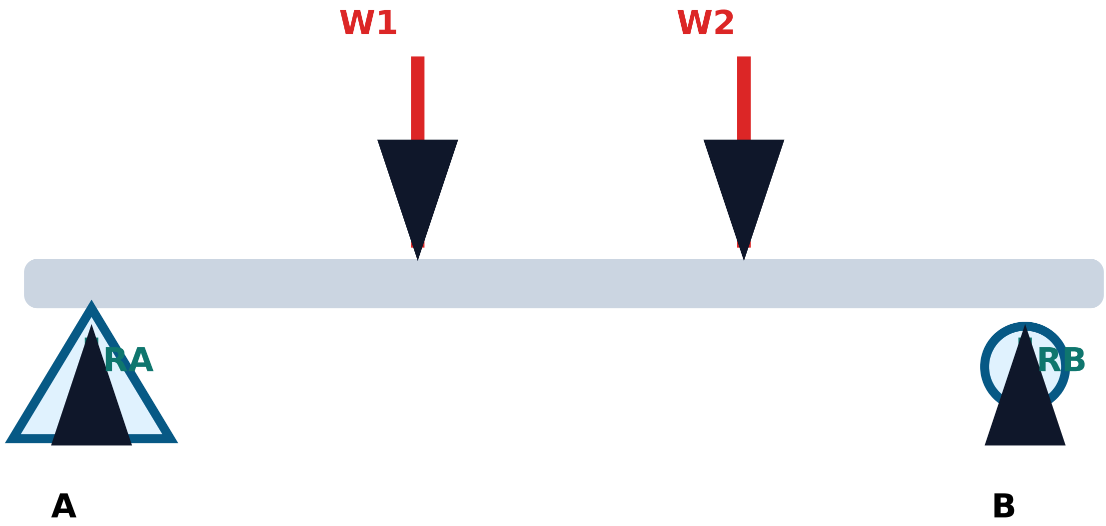
$$\Sigma V = 0 \Rightarrow RA + RB = \text{total downward load}$$

$$\Sigma MA = 0 \Rightarrow RB \times \text{span} = \text{sum of load moments about A}$$

After finding RB, find RA from vertical equilibrium. Experimental readings are compared with theoretical values.

Beam reaction problem-□ □□□□□ diagram □□□□□□□□□□. Load □□□□□ acting □□□□□□□□□□ □□□□□ distance □□□□□ mark □□□□□□□□ calculation □□□□□□□□□□.

Beam reaction diagram



M2.02 Support reactions using graphical method by drawing link polygon

The graphical method uses a space diagram, load line, pole and link polygon. The intersection of first and last link gives reaction location. This method trains drawing accuracy and visual understanding of equilibrium.

What to draw

- Space diagram with beam, supports and loads.
- Force polygon/load line to scale.
- Pole point and rays.
- Link polygon from the loading diagram.

Practical application

Beam reaction calculation is used before selecting bearings, supports, base plates or brackets. In an escalator/elevator plant, similar logic is used while checking support positions, landing loads, machine base and maintenance platforms.

Troubleshooting

- If measured reactions do not add up to total load, check zero error and support friction.
- If one reaction becomes negative in calculation, load position/support assumption is wrong or overhanging case is involved.

Module 2 expected questions

- Define support reaction.
- Draw a simply supported beam with point loads.

- Derive reactions using moment equation.
- Explain link polygon method.
- Why is percentage error calculated in lab?

Common mistakes

- Moment taken about wrong point or wrong unit.
- Using mm in one place and metre in another place.
- Forgetting self-weight if apparatus instruction includes it.
- Reporting answer without unit.

Module 3 • CO3 • 9 Hours

Centroid, Moment of Inertia and Bell Crank Lever

Outcome: infer centre of gravity and mass moment of inertia. This module connects static balance with rotating-system behaviour.

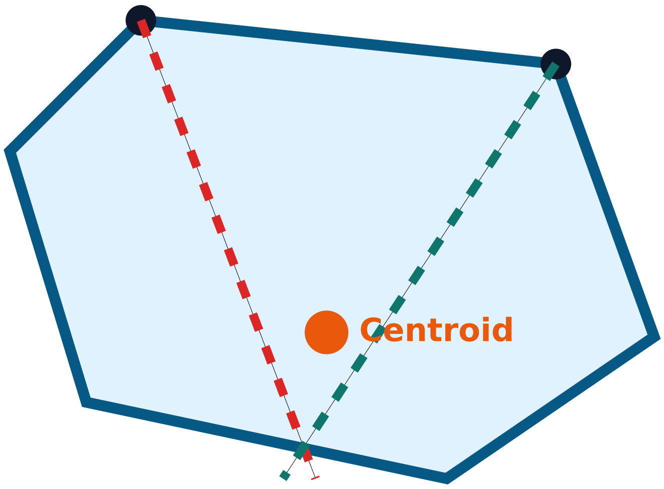
M3.01 Determination of centroid of different laminae

The centroid is the geometric centre of an area. For a uniform thin lamina, the centre of gravity coincides with the centroid. Suspend the lamina from one hole and mark the vertical line using a plumb bob. Suspend from another hole and repeat. The intersection gives the centroid.

$$\bar{x} = \Sigma(Ax) / \Sigma A, \quad \bar{y} = \Sigma(Ay) / \Sigma A$$

For irregular laminae, the suspension method is more practical than calculation.

Centroid by suspension



M3.02 Mass moment of inertia of flywheel and shaft

Mass moment of inertia is the resistance of a rotating body to change in angular motion. In flywheel experiment, energy supplied by a falling mass is related to rotational kinetic energy and work lost in friction. Readings include mass, fall height, number of rotations during fall and number of rotations after detachment.

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Kinetic energy = 1/2 Iω²
```

Use consistent SI units. The final result is normally expressed in $kg \cdot m^2$.

M3.03 Verify law of moments using bell crank lever

The law of moments states that for equilibrium, clockwise moments equal anticlockwise moments about a point. In bell crank lever apparatus, loads are applied at different arms and the balance condition is checked.

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Clockwise moment = Anticlockwise moment
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F1 × d1 = F2 × d2
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Module 3 expected questions

- Define centroid and centre of gravity.

- Explain plumb line method for lamina.
- Define mass moment of inertia.
- State law of moments.
- Write two applications of flywheel.

Industrial note

Centroid helps in lifting plates safely and designing balanced components. Moment of inertia matters in motors, flywheels, shafts and rotating machine parts. The bell crank lever principle is used in linkages, brakes, valves and mechanical actuators.

> Moment = force × perpendicular distance. Distance □□□□□□□□□ □□□ force □□□□□□ □□□□□□ turning effect □□□□□□□□.

Module 4 • CO4 • 11 Hours + Lab Exam II

Strain and Coefficient of Friction

Outcome: determine strain under axial tension and coefficient of friction on horizontal and inclined planes through experimentation.

M4.01 Determination of strain under tensile loading

Strain is deformation per unit original dimension. Under axial tensile load, the specimen extends along the loading direction and may contract in the perpendicular direction. Strain has no unit because it is a ratio.

$$\text{Longitudinal strain} = \Delta L / L$$

$$\text{Lateral strain} = \Delta d / d$$

Use a proper gauge length, take repeated readings and avoid sudden loading.

M4.02 Coefficient of friction on horizontal plane

For a block on a horizontal surface, gradually increase pulling force until impending motion. At limiting condition, the coefficient of friction is:

$$\mu = F / N$$

where F is limiting friction or pull at start of motion and N is normal reaction. On a horizontal plane, N is approximately equal to weight of the block plus added load.

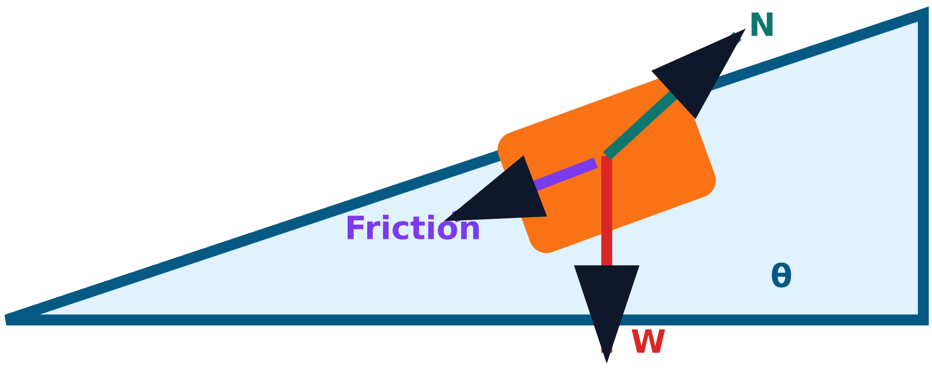
M4.03 Coefficient of friction on inclined plane

Increase the angle of inclination slowly until the block just starts sliding. This angle is the angle of repose. At that condition:

$$\mu = \tan \theta$$

The surface must be clean and dry unless the experiment specifically asks for another condition.

Friction plane diagram



Common mistakes

- Reading angle before the block just starts moving.
- Applying jerk instead of gradual pull.
- Not cleaning surface before friction test.
- Confusing stress and strain.

Module 4 expected questions

- Define strain and write formula.
- Explain coefficient of friction.
- Derive $\mu = \tan\theta$ for inclined plane at limiting condition.
- Write precautions in tensile loading experiment.
- Give applications of friction in machines.

Formula / Rule Bank

Use these formulas with variables, units and one practical application example.

Resultant of two forces

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

P,Q are forces in newton; θ is included angle. Used for two concurrent forces.

Example: two pulls on a hook combine into one equivalent pull.

Direction of resultant

$$\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$$

α is angle between resultant and force P. Used in vector resultant problems.

Lami’s theorem

$$F1/\sin\alpha = F2/\sin\beta = F3/\sin\gamma$$

Used for three coplanar concurrent forces in equilibrium, such as Jib crane.

Beam equilibrium

$$\Sigma V = 0,\ \Sigma M = 0$$

Used to find support reactions. Moment unit is N·m or N·mm.

Moment

$$M = F \times d$$

F in N, d in m; moment in N·m. Used in bell crank lever and beam reactions.

Centroid of composite area

$$\bar{x} = \Sigma Ax / \Sigma A,\ \bar{y} = \Sigma Ay / \Sigma A$$

A is area, x/y are distances from reference axes. Used for laminae.

Rotational kinetic energy

$$KE = 1/2\ I\omega^2$$

Strain

$$\varepsilon = \Delta L / L$$

I in kg·m²; ω in rad/s. Used in flywheel inertia experiment.

Dimensionless ratio. Used in tensile loading experiment.

Horizontal friction

$$\mu = F / N$$

F is limiting friction, N is normal reaction. Used to compare surfaces.

Inclined plane friction

$$\mu = \tan\theta$$

θ is angle of repose. Used when block just begins to slide.

Solved Examples / Problem Lab

Do not write final answers only. Show data, formula, substitution, unit and conclusion.

Example 1 - Beam reaction

Given: Simply supported beam span 4 m. Load 200 N at 1 m from A and 300 N at 3 m from A.

Formula: $\Sigma M_A=0 \Rightarrow R_B\times4 = 200\times1 + 300\times3 = 1100$
RB: $1100/4 = 275$ N
RA: total load - RB = 500 - 275 = 225 N
Answer: RA = 225 N, RB = 275 N upward.

Exam tip: show beam diagram before calculation.

Example 2 - Coefficient of friction

Given: A block weighing 50 N starts moving when horizontal pull is 18 N.

Formula: $\mu = F/N$
Substitution: $\mu = 18/50 = 0.36$
Answer: coefficient of friction = 0.36.

Example 3 - Inclined plane

Given: A block starts sliding at $\theta = 28°$.

Formula: $\mu = \tan\theta$
Substitution: $\mu = \tan 28° = 0.532$
Answer: $\mu \approx 0.53$.

Example 4 - Strain

Given: Original length = 200 mm, extension = 0.24 mm.

Formula: $\varepsilon = \Delta L/L$
Substitution: $\varepsilon = 0.24/200 = 0.0012$
Answer: strain = 1.2×10^{-3}. No unit.

Example 5 - Bell crank lever

Example 6 - Percentage error

Given: Force F1 = 80 N acts at arm 0.15 m. Find balancing force at arm 0.24 m.

Law: $F_1 d_1 = F_2 d_2$

Substitution: $80 \times 0.15 = F_2 \times 0.24$

F2: $12/0.24 = 50$ N

Answer: balancing force = 50 N.

Given: theoretical reaction = 275 N, experimental reaction = 268 N.

Formula: error % = |theoretical - experimental| / theoretical × 100

Substitution: $|275-268|/275 \times 100 = 2.55\%$

Answer: percentage error = 2.55%.

Practice Section and Quick Revision

Use this section for last-day revision, viva and lab record checking.

One-line revision points

- Resultant replaces a force system by one equivalent force.
- Equilibrant balances the resultant and acts in opposite direction.
- Moment is force multiplied by perpendicular distance.
- Beam reactions are found using equilibrium equations.
- Centroid of uniform lamina is its centre of gravity.
- Moment of inertia measures resistance to angular acceleration.
- Strain is change in length divided by original length.
- Coefficient of friction is limiting friction divided by normal reaction.

Common exam traps

- Leaving observation table incomplete.
- Forgetting unit conversion.
- Not drawing neat apparatus diagram.
- Using theoretical answer as experimental result.
- Writing conclusion without comparing error.

Objective questions

No.	Question	Options
1	The SI unit of force is	A. joule B. watt C. newton D. pascal
2	Moment is calculated by	A. F/d B. F×d C. d/F D. F+d
3	The force that balances resultant is	A. reaction B. equilibrant C. moment D. strain
4	For a horizontal surface, normal reaction is approximately equal to	A. pull B. weight C. friction D. angle

No.	Question	Options
5	At angle of repose, μ equals	A. sinθ B. cosθ C. tanθ D. cotθ
6	Strain is	A. force/area B. extension/original length C. weight/volume D. moment/radius
7	A beam with pin and roller support is	A. cantilever B. fixed beam C. simply supported beam D. shaft
8	Bell crank lever verifies	A. law of moments B. Ohm's law C. Faraday's law D. Hooke's law only
9	Point through which body weight acts is	A. centre of gravity B. pole C. link D. pulley
10	Force table pulley should be smooth to reduce	A. voltage drop B. friction error C. corrosion D. heat loss

Answer key: 1-C, 2-B, 3-B, 4-B, 5-C, 6-B, 7-C, 8-A, 9-A, 10-B.

Fresh Model Practical Question Paper

The supplied syllabus mentions lab exams but does not give a detailed official model paper pattern. This paper is freshly created for practice from the official experiment list.

Engineering Mechanics Lab - Model Practical Examination

Course code: 2029 **Time:** 3 Hours **Maximum marks:** 50 practice marks

Instruction: Draw neat diagrams, write observations with units, show formula and calculation, state result and answer viva questions orally.

Q.No.	Part	Question	Marks
1	Part A - Apparatus	Identify any five apparatus used in Engineering Mechanics Lab and write one use of each.	5
2	Part A - Viva	Define equilibrant, state Lami’s theorem, define coefficient of friction, state law of moments and define strain.	5
3	Part B - Main experiment	Determine support reactions of a simply supported beam carrying two point loads. Draw diagram, record readings, calculate reactions and compare with experiment.	20
4	Part B - Internal choice	OR Determine resultant of a concurrent force system by graphical method and verify with force table experiment.	20

Q.No.	Part	Question	Marks
5	Part C	Prepare observation table, sample calculation, percentage error and result statement.	10
6	Part D	Write precautions, sources of error, industrial application and final conclusion.	10

Complete Model Answers

These are full answer guides. In actual lab, numerical values must be replaced by observed readings.

Q1 answer

List apparatus such as force table, pulley, slotted weights, weight hanger, spring balance, simply supported beam apparatus, plumb bob, drawing board, set square, bell crank lever, flywheel apparatus and friction plane. Each use must be specific.

Q2 answer

Equilibrant is equal and opposite to resultant. Lami’s theorem: $F1/\sin\alpha = F2/\sin\beta = F3/\sin\gamma$. Coefficient of friction is limiting friction divided by normal reaction. Law of moments: clockwise moments equal anticlockwise moments in equilibrium. Strain is change in dimension divided by original dimension.

Q3 answer

Draw beam with span, support A/B and load positions. Record load values and distances. Use $\Sigma V=0$ and $\Sigma M=0$. Take moment about A to find RB, then RA = total load - RB. Compare with experimental reactions and calculate percentage error.

Q4 answer

Choose scale, draw forces head-to-tail and measure closing side as resultant. In force table, set known forces, find equilibrant by keeping centre ring at centre, then resultant is equal and opposite to equilibrant. Include vector polygon and observation table.

Q5 answer

Observation table must include load, distance/angle, trial readings, average value, calculated value, experimental value and percentage error. Formula: |theoretical - experimental| / theoretical $\times$ 100.

Q6 answer

Precautions: avoid jerks, check zero error, use correct scale, avoid pulley friction, keep surfaces clean, take repeated readings and handle weights safely. Applications: beam reactions for structures, friction for brakes/guides, strain for material testing and resultant for crane/rigging analysis.

References

Official references from syllabus: Khurmi Applied Mechanics; Bansal Engineering Mechanics; Ramamrutham Engineering Mechanics; Bedi Strength of Materials; Khurmi Strength of Materials; Ramamurtham and Punmia Strength of Materials; Ram & Chauhan Applied Mechanics; Meriam & Kraige Statics; Rattan Strength of Materials; Subramaniam Strength of Materials.

Online resources listed: NPTEL 122104015, NPTEL 112103109 and Virtual Labs.

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